

CAPITAL CITY, PA, USA

WMO: 725118

Lat: 40.217N

Lon: 76.851W

Elev: 340

StdP: 14.52

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 14751

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.0	(c) 15.5	(d) -4.9	(e) 4.3	(f) 14.8	(g) -0.8	(h) 5.3	(i) 18.3	(j) 24.5	(k) 31.9	(l) 22.0	(m) 31.4	(n) 8.0	(o) 290	(p) 0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	92.2	74.2	89.7	72.8	87.3	71.7	76.7	87.5	75.4	85.1	74.2	83.2	7.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.3	125.5	81.3	72.3	121.3	80.2	71.5	117.6	79.2	40.2	87.3	38.9	85.7	37.8	83.3	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 20.5	(b) 18.0	(c) 15.9	(e) 6.1	(f) 96.3	(g) 4.7	(h) 3.3	(i) 2.7	(j) 98.7	(k) 0.0	(l) 100.6	(m) -2.7	(n) 102.5	(o) -6.1	(p) 104.9		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			6.1	96.3	4.7	3.3	2.7	98.7	0.0	100.6	-2.7	102.5	-6.1	104.9		
			4.4	79.6	4.3	1.4	1.3	80.6	-1.2	81.4	-3.6	82.2	-6.7	83.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	31.1	33.3	41.8	53.7	63.4	71.9	76.5	74.9	68.1	56.3	45.3	36.0
	DBStd	17.73	9.51	8.95	9.31	8.62	7.77	5.97	5.00	5.03	6.58	8.33	7.90	8.30
	HDD50	2061	588	471	289	55	2	0	0	0	0	31	184	441
	HDD65	4974	1050	886	721	352	127	13	0	1	41	292	591	900
	CDD50	3697	3	5	34	166	419	658	820	773	543	227	43	6
	CDD65	1136	0	0	0	13	79	221	356	309	135	23	0	0
	CDH74	10026	0	3	12	204	717	1936	3374	2601	1023	154	2	0
	CDH80	3459	0	0	0	52	206	643	1318	895	316	29	0	0
Wind	WSAvg	6.3	7.1	7.3	7.8	7.5	6.2	5.9	5.3	4.9	5.0	5.8	6.2	6.6
Precipitation	PrecAvg	40.7	2.8	2.9	3.3	3.2	4.4	3.8	3.6	3.3	3.5	2.9	3.5	3.2
	PrecMax	59.3	8.0	5.9	6.1	8.0	9.7	18.5	9.7	7.8	15.0	9.9	7.2	7.6
	PrecMin	30.1	0.4	0.6	1.0	0.4	0.5	0.1	0.8	0.9	0.6	0.0	0.8	0.8
	PrecStd	6.9	1.7	1.4	1.3	1.6	2.2	3.3	2.0	1.6	2.9	1.8	1.6	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.3	68.1	75.2	85.9	89.6	93.0	96.5	94.8	91.9	83.7	72.3	65.3
		MCWB	56.0	54.8	57.9	64.8	71.3	74.5	75.7	75.5	72.7	70.4	60.4	58.8
	2%	DB	55.0	57.8	67.9	80.5	85.4	89.8	92.9	90.9	87.7	78.6	66.9	57.3
		MCWB	48.5	48.2	55.4	62.9	69.2	72.9	74.9	73.8	71.3	66.7	58.0	51.8
	5%	DB	48.4	50.8	62.3	74.5	81.8	86.6	89.8	88.0	83.4	73.9	62.9	52.1
		MCWB	43.6	42.3	52.2	59.4	67.3	71.1	73.2	72.6	69.3	64.4	54.9	47.4
	10%	DB	44.6	45.7	57.0	69.7	77.9	83.7	87.2	84.9	80.1	70.2	58.6	47.9
		MCWB	40.0	39.7	48.1	56.5	65.2	69.4	72.1	71.3	67.4	61.8	51.5	43.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.3	59.0	61.4	68.6	74.2	76.6	78.8	78.3	76.0	72.5	63.9	60.1
		MCDB	62.1	66.4	69.7	80.9	85.1	87.6	91.8	88.8	87.4	79.4	68.2	64.4
	2%	WB	48.9	49.3	58.0	65.0	71.5	75.0	76.9	76.7	73.8	69.7	60.1	53.8
		MCDB	53.7	56.6	65.8	75.9	81.5	85.6	88.5	86.3	81.9	75.9	65.3	56.4
	5%	WB	44.4	44.0	53.5	62.0	69.5	73.4	75.4	75.1	72.1	66.5	56.3	47.8
		MCDB	48.0	49.4	60.6	71.6	78.7	83.2	85.6	83.6	79.5	72.5	61.2	51.3
	10%	WB	40.3	40.3	49.1	58.8	67.5	72.0	74.2	73.7	70.4	62.8	52.7	43.8
		MCDB	43.9	45.6	56.0	67.6	75.2	80.6	83.5	81.7	77.0	68.3	57.1	47.4
Mean Daily Temperature Range	5% DB	MDBR	14.1	15.6	17.7	20.0	19.3	18.7	18.6	17.9	18.6	17.8	16.7	13.6
		MCDBR	20.5	24.1	25.7	28.7	24.2	21.8	21.0	20.7	22.7	23.2	23.4	19.1
	5% WB	MCWBR	16.3	16.7	16.4	14.8	11.1	9.0	8.0	7.9	9.5	12.5	15.4	15.9
		MCDWR	19.5	21.0	22.9	24.4	21.2	19.0	18.0	17.6	17.5	18.3	19.1	17.2
Clear-Sky Solar Irradiance	taub		0.312	0.330	0.368	0.400	0.430	0.502	0.513	0.501	0.420	0.372	0.329	0.310
	taud		2.424	2.349	2.324	2.267	2.216	2.050	2.033	2.052	2.251	2.392	2.477	2.469
	Ebn at Noon		269	280	280	277	270	250	246	245	259	261	261	261
	Edh at Noon		26	32	37	42	45	53	53	51	39	30	24	23
All-Sky Solar Radiation	RadAvg		574	859	1149	1490	1672	1853	1867	1635	1336	948	666	488
	RadStd		48	108	84	139	194	154	103	90	175	107	64	57

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+365	+207
(47) Regional (64 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+214	+117

Nomenclature: See separate page